

```

# !pip install torchvision

import torch
import os
from torch.utils.data import DataLoader, Dataset
from torchvision import transforms
from PIL import Image

# images load => transform => dataset of all imgs
class ImageProcessor:
    def __init__(self, root_dir_path, transformations=None):
        self.root_dir_path = root_dir_path
        self.transformations = transformations

        # list of paths for all images
        self.all_img_paths = [os.path.join(root_dir_path, img) for img
in os.listdir(root_dir_path)]

    def __len__(self):
        return len(self.all_img_paths)

    def __getitem__(self, idx):
        img_path = self.all_img_paths[idx]
        img = Image.open(img_path).convert("RGB")

        if self.transformations:
            img = self.transformations(img)

        return img

root_dir_path = "./img_align_celeba"

transformations = transforms.Compose([
    transforms.CenterCrop(178), # 178x218 => 178x178
    transforms.Resize(64), # 64x64
    transforms.ToTensor(),
    transforms.Normalize((0.5, 0.5, 0.5), (0.5, 0.5, 0.5)) # [-1, 1]
])

dataset = ImageProcessor(root_dir_path, transformations)
print(f"loaded {len(dataset)} images")

loaded 202599 images

dataloader = DataLoader(dataset, batch_size=128, shuffle=True)

```

Generator Network

```

import torch.nn as nn
import torch.optim as optim
import numpy as np

```

```

class Generator(nn.Module):
    def __init__(self, z_dim=100, img_channels=3): # 3 is for RGB
        super(Generator, self).__init__()

        # fully connected (dense) layers
        self.model = nn.Sequential(
            nn.Linear(z_dim, 256), # 100 => 256
            nn.ReLU(),

            nn.Linear(256, 512),
            nn.ReLU(),

            nn.Linear(512, 1024),
            nn.ReLU(),

            nn.Linear(1024, 64 * 64 * img_channels),
            nn.Tanh() # [-1, 1]
        )

    def forward(self, z):
        img = self.model(z)
        img = img.view(img.size(0), 3, 64, 64)
        return img

    # fake img => 64 x 64 x 3 x batch_size

```

Discriminator Network

```

class Discriminator(nn.Module):
    def __init__(self, img_channels=3): # 3 is for RGB
        super(Discriminator, self).__init__()

        # fully connected (dense) layers
        self.model = nn.Sequential(
            nn.Flatten(), # 4D tensor => 1D

            nn.Linear(img_channels * 64 * 64, 1024),
            nn.LeakyReLU(0.2, inplace=True),

            nn.Linear(1024, 512),
            nn.LeakyReLU(0.2, inplace=True),

            nn.Linear(512, 256),
            nn.LeakyReLU(0.2, inplace=True),

            nn.Linear(256, 1),
            nn.Sigmoid() # probability of being real/fake
        )

```

```

    def forward(self, img):
        return self.model(img)

GAN_loss = nn.BCELoss()

generator = Generator()
g_optimizer = optim.Adam(generator.parameters(), lr=0.0002,
betas=(0.5, 0.999))

discriminator = Discriminator()
d_optimizer = optim.Adam(discriminator.parameters(), lr=0.0002,
betas=(0.5, 0.999))

import torch
# device
if torch.backends.mps.is_available():
    device = torch.device("mps")
elif torch.cuda.is_available():
    device = torch.device("cuda")
else:
    device = torch.device("cpu")

print(f"device is {device}")

device is mps

generator = generator.to(device)
discriminator = discriminator.to(device)

```

Training the GAN

```

def train(generator, discriminator, dataloader, epochs=10):
    for epoch in range(epochs):
        for i, imgs in enumerate(dataloader):
            real_imgs = imgs.to(device)
            batch_size = real_imgs.size(0)

            # creates real imgs labels & fake imgs labels
            real_labels = torch.ones(batch_size, 1).to(device) # [1,
1, 1....]
            fake_labels = torch.zeros(batch_size, 1).to(device) # [0,
0, 0....]

            # Train the Discriminator
            d_optimizer.zero_grad()

            fake_imgs = generator(torch.randn(batch_size,
100).to(device))

            real_loss = GAN_loss(discriminator(real_imgs),

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real_labels)
    fake_loss = GAN_loss(discriminator(fake_imgs.detach()),
fake_labels)

    d_loss = (real_loss + fake_loss) / 2

    d_loss.backward()
    d_optimizer.step()

    # Train the Generator
    g_optimizer.zero_grad()

    g_loss = GAN_loss(discriminator(fake_imgs), real_labels)

    g_loss.backward()
    g_optimizer.step()

    if i % 50 == 0:
        print(f"for epoch: {epoch+1}/{epochs}... batch:
{i+1}... G-loss:{g_loss}.... D-loss: {d_loss}")

        # save generated imgs for each epoch
        save_generated_images(generator, epoch, device)

import matplotlib.pyplot as plt
import torchvision

def save_generated_images(generator, epoch, device, num_imgs=8):
    z = torch.randn(num_imgs, 100).to(device)
    generated_imgs = generator(z).detach().cpu()

    grid = torchvision.utils.make_grid(generated_imgs, nrow=4,
normalize=True)

    plt.imshow(np.transpose(grid, (1, 2, 0)))
    plt.title(f"epoch {epoch+1}")
    plt.axis("off")
    plt.show()

train(generator, discriminator, dataloader, epochs=5)

for epoch: 1/5... batch: 1... G-loss:3.322542190551758.... D-loss:
0.3139531910419464
for epoch: 1/5... batch: 51... G-loss:2.3338780403137207.... D-loss:
0.33996284008026123
for epoch: 1/5... batch: 101... G-loss:1.9056925773620605.... D-loss:
0.27508699893951416
for epoch: 1/5... batch: 151... G-loss:2.290879726409912.... D-loss:
0.31506916880607605
for epoch: 1/5... batch: 201... G-loss:3.2655439376831055.... D-loss:
0.38347920775413513

```

for epoch: 1/5... batch: 251... G-loss:1.425288200378418.... D-loss:
0.4764195680618286
for epoch: 1/5... batch: 301... G-loss:2.7879319190979004.... D-loss:
0.3383023738861084
for epoch: 1/5... batch: 351... G-loss:2.234636068344116.... D-loss:
0.5133187174797058
for epoch: 1/5... batch: 401... G-loss:3.0106489658355713.... D-loss:
0.3158843517303467
for epoch: 1/5... batch: 451... G-loss:2.420163154602051.... D-loss:
0.33349305391311646
for epoch: 1/5... batch: 501... G-loss:2.933938980102539.... D-loss:
0.40808793902397156
for epoch: 1/5... batch: 551... G-loss:2.3494484424591064.... D-loss:
0.46661049127578735
for epoch: 1/5... batch: 601... G-loss:1.683651328086853.... D-loss:
0.41399064660072327
for epoch: 1/5... batch: 651... G-loss:2.7929182052612305.... D-loss:
0.4224250912666321
for epoch: 1/5... batch: 701... G-loss:2.230024814605713.... D-loss:
0.36520570516586304
for epoch: 1/5... batch: 751... G-loss:2.260885715484619.... D-loss:
0.3514847159385681
for epoch: 1/5... batch: 801... G-loss:2.148690938949585.... D-loss:
0.43477708101272583
for epoch: 1/5... batch: 851... G-loss:3.2518835067749023.... D-loss:
0.36269596219062805
for epoch: 1/5... batch: 901... G-loss:2.445046901702881.... D-loss:
0.32137617468833923
for epoch: 1/5... batch: 951... G-loss:1.8141160011291504.... D-loss:
0.4729716181755066
for epoch: 1/5... batch: 1001... G-loss:2.2243471145629883.... D-loss:
0.392625629901886
for epoch: 1/5... batch: 1051... G-loss:2.242885112762451.... D-loss:
0.38423967361450195
for epoch: 1/5... batch: 1101... G-loss:1.534072756767273.... D-loss:
0.494789719581604
for epoch: 1/5... batch: 1151... G-loss:1.9877220392227173.... D-loss:
0.41874587535858154
for epoch: 1/5... batch: 1201... G-loss:1.8833907842636108.... D-loss:
0.4584863483905792
for epoch: 1/5... batch: 1251... G-loss:1.9756325483322144.... D-loss:
0.5721912384033203
for epoch: 1/5... batch: 1301... G-loss:2.1495420932769775.... D-loss:
0.3360183835029602
for epoch: 1/5... batch: 1351... G-loss:1.563887596130371.... D-loss:
0.5648462176322937
for epoch: 1/5... batch: 1401... G-loss:1.9550179243087769.... D-loss:
0.4255979061126709
for epoch: 1/5... batch: 1451... G-loss:2.1915459632873535.... D-loss:

0.39103826880455017
for epoch: 1/5... batch: 1501... G-loss:2.6412224769592285.... D-loss:
0.3637201189994812
for epoch: 1/5... batch: 1551... G-loss:3.0327584743499756.... D-loss:
0.4363042116165161

epoch 1



for epoch: 2/5... batch: 1... G-loss:2.224043130874634.... D-loss:
0.3620590269565582
for epoch: 2/5... batch: 51... G-loss:1.9637258052825928.... D-loss:
0.3923010230064392
for epoch: 2/5... batch: 101... G-loss:2.2164976596832275.... D-loss:
0.39461421966552734
for epoch: 2/5... batch: 151... G-loss:2.699800968170166.... D-loss:
0.41392046213150024
for epoch: 2/5... batch: 201... G-loss:1.8283586502075195.... D-loss:
0.3859618902206421
for epoch: 2/5... batch: 251... G-loss:2.3580565452575684.... D-loss:
0.4057340621948242
for epoch: 2/5... batch: 301... G-loss:1.7985527515411377.... D-loss:
0.48905372619628906
for epoch: 2/5... batch: 351... G-loss:1.6815409660339355.... D-loss:
0.4434852600097656
for epoch: 2/5... batch: 401... G-loss:1.9040369987487793.... D-loss:
0.3765515685081482
for epoch: 2/5... batch: 451... G-loss:2.185502529144287.... D-loss:
0.44115644693374634
for epoch: 2/5... batch: 501... G-loss:2.062896251678467.... D-loss:
0.4228114187717438
for epoch: 2/5... batch: 551... G-loss:1.924088478088379.... D-loss:
0.4383426904678345
for epoch: 2/5... batch: 601... G-loss:1.3971140384674072.... D-loss:
0.46737000346183777

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for epoch: 2/5... batch: 651... G-loss:1.8383278846740723.... D-loss:
0.434894859790802
for epoch: 2/5... batch: 701... G-loss:1.7875157594680786.... D-loss:
0.49363288283348083
for epoch: 2/5... batch: 751... G-loss:2.079268217086792.... D-loss:
0.4369892477989197
for epoch: 2/5... batch: 801... G-loss:2.1218652725219727.... D-loss:
0.37581923604011536
for epoch: 2/5... batch: 851... G-loss:2.421546459197998.... D-loss:
0.562237024307251
for epoch: 2/5... batch: 901... G-loss:2.377397060394287.... D-loss:
0.513087809085846
for epoch: 2/5... batch: 951... G-loss:1.5487229824066162.... D-loss:
0.4786056876182556
for epoch: 2/5... batch: 1001... G-loss:1.4613673686981201.... D-loss:
0.44867855310440063
for epoch: 2/5... batch: 1051... G-loss:1.9650614261627197.... D-loss:
0.47719258069992065
for epoch: 2/5... batch: 1101... G-loss:2.2828028202056885.... D-loss:
0.4135931730270386
for epoch: 2/5... batch: 1151... G-loss:2.1250555515289307.... D-loss:
0.44256144762039185
for epoch: 2/5... batch: 1201... G-loss:1.8080735206604004.... D-loss:
0.41960299015045166
for epoch: 2/5... batch: 1251... G-loss:2.1919941902160645.... D-loss:
0.46693187952041626
for epoch: 2/5... batch: 1301... G-loss:1.6531965732574463.... D-loss:
0.4606369435787201
for epoch: 2/5... batch: 1351... G-loss:1.9671275615692139.... D-loss:
0.4479203224182129
for epoch: 2/5... batch: 1401... G-loss:2.5748839378356934.... D-loss:
0.47587645053863525
for epoch: 2/5... batch: 1451... G-loss:1.7222375869750977.... D-loss:
0.44513243436813354
for epoch: 2/5... batch: 1501... G-loss:1.983347773551941.... D-loss:
0.5210983753204346
for epoch: 2/5... batch: 1551... G-loss:2.145564556121826.... D-loss:
0.4665418267250061
```

epoch 2



```
for epoch: 3/5... batch: 1... G-loss:1.6577541828155518.... D-loss:
0.5077065229415894
for epoch: 3/5... batch: 51... G-loss:1.9990367889404297.... D-loss:
0.39551958441734314
for epoch: 3/5... batch: 101... G-loss:1.504288911819458.... D-loss:
0.592709481716156
for epoch: 3/5... batch: 151... G-loss:1.866033673286438.... D-loss:
0.44140589237213135
for epoch: 3/5... batch: 201... G-loss:1.635092854499817.... D-loss:
0.5809305906295776
for epoch: 3/5... batch: 251... G-loss:1.5634510517120361.... D-loss:
0.5191962718963623
for epoch: 3/5... batch: 301... G-loss:2.1973390579223633.... D-loss:
0.4810156524181366
for epoch: 3/5... batch: 351... G-loss:2.5110936164855957.... D-loss:
0.5778912305831909
for epoch: 3/5... batch: 401... G-loss:1.808459997177124.... D-loss:
0.4934077262878418
for epoch: 3/5... batch: 451... G-loss:1.719606637954712.... D-loss:
0.45575809478759766
for epoch: 3/5... batch: 501... G-loss:1.7246140241622925.... D-loss:
0.4646170735359192
for epoch: 3/5... batch: 551... G-loss:1.5188372135162354.... D-loss:
0.4623265564441681
for epoch: 3/5... batch: 601... G-loss:2.123291015625.... D-loss:
0.480472594499588
for epoch: 3/5... batch: 651... G-loss:1.4162521362304688.... D-loss:
0.5428123474121094
for epoch: 3/5... batch: 701... G-loss:1.6925063133239746.... D-loss:
0.5116690397262573
for epoch: 3/5... batch: 751... G-loss:1.7364311218261719.... D-loss:
0.5039415955543518
```



```
for epoch: 3/5... batch: 801... G-loss:1.7333842515945435.... D-loss:
0.46812182664871216
for epoch: 3/5... batch: 851... G-loss:1.5875780582427979.... D-loss:
0.4062122404575348
for epoch: 3/5... batch: 901... G-loss:2.004241943359375.... D-loss:
0.5951343178749084
for epoch: 3/5... batch: 951... G-loss:1.585517168045044.... D-loss:
0.44435375928878784
for epoch: 3/5... batch: 1001... G-loss:1.652601718902588.... D-loss:
0.5118982791900635
for epoch: 3/5... batch: 1051... G-loss:1.920764684677124.... D-loss:
0.5785634517669678
for epoch: 3/5... batch: 1101... G-loss:1.7606570720672607.... D-loss:
0.46966680884361267
for epoch: 3/5... batch: 1151... G-loss:1.8862855434417725.... D-loss:
0.5157556533813477
for epoch: 3/5... batch: 1201... G-loss:1.3033509254455566.... D-loss:
0.5193349123001099
for epoch: 3/5... batch: 1251... G-loss:1.6174163818359375.... D-loss:
0.4986237585544586
for epoch: 3/5... batch: 1301... G-loss:1.5013707876205444.... D-loss:
0.45492804050445557
for epoch: 3/5... batch: 1351... G-loss:2.3075318336486816.... D-loss:
0.46571147441864014
for epoch: 3/5... batch: 1401... G-loss:1.6374523639678955.... D-loss:
0.5487385988235474
for epoch: 3/5... batch: 1451... G-loss:1.784045696258545.... D-loss:
0.5606552362442017
for epoch: 3/5... batch: 1501... G-loss:1.6892889738082886.... D-loss:
0.4774284064769745
for epoch: 3/5... batch: 1551... G-loss:2.4674465656280518.... D-loss:
0.49953678250312805
```

epoch 3



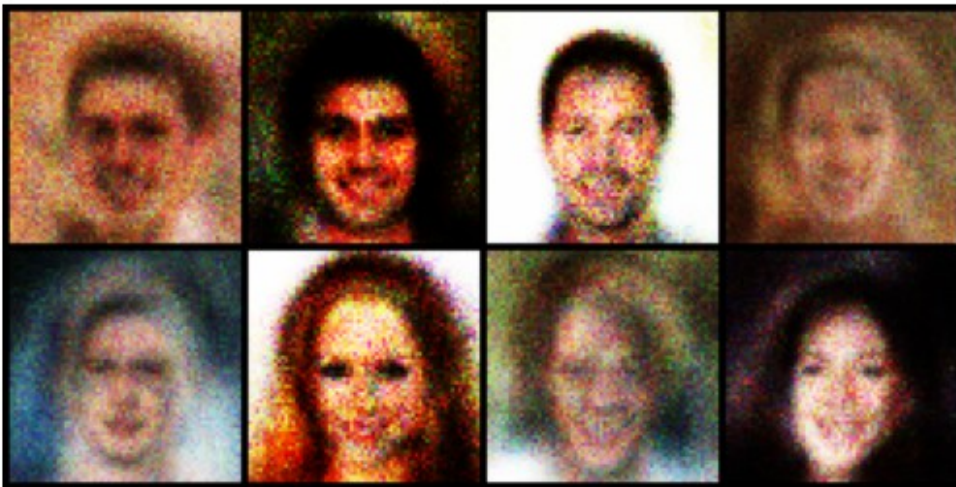
```
for epoch: 4/5... batch: 1... G-loss:1.2645459175109863.... D-loss:
0.57386714220047
for epoch: 4/5... batch: 51... G-loss:1.760229468345642.... D-loss:
0.480736643075943
for epoch: 4/5... batch: 101... G-loss:1.5768831968307495.... D-loss:
0.5596146583557129
for epoch: 4/5... batch: 151... G-loss:1.5744142532348633.... D-loss:
0.5647860765457153
for epoch: 4/5... batch: 201... G-loss:2.049656391143799.... D-loss:
0.5023221373558044
for epoch: 4/5... batch: 251... G-loss:1.7594491243362427.... D-loss:
0.555428147315979
for epoch: 4/5... batch: 301... G-loss:1.6680982112884521.... D-loss:
0.5410287380218506
for epoch: 4/5... batch: 351... G-loss:1.5169036388397217.... D-loss:
0.5803616046905518
for epoch: 4/5... batch: 401... G-loss:1.653578519821167.... D-loss:
0.46837514638900757
for epoch: 4/5... batch: 451... G-loss:1.687021255493164.... D-loss:
0.5012955665588379
for epoch: 4/5... batch: 501... G-loss:1.5630114078521729.... D-loss:
0.5092208385467529
for epoch: 4/5... batch: 551... G-loss:1.6734384298324585.... D-loss:
0.49534159898757935
for epoch: 4/5... batch: 601... G-loss:1.889878273010254.... D-loss:
0.45303308963775635
for epoch: 4/5... batch: 651... G-loss:1.5882595777511597.... D-loss:
0.6346449851989746
for epoch: 4/5... batch: 701... G-loss:1.6632155179977417.... D-loss:
0.45915505290031433
for epoch: 4/5... batch: 751... G-loss:1.507645845413208.... D-loss:
0.49515992403030396
for epoch: 4/5... batch: 801... G-loss:1.388528823852539.... D-loss:
0.5422342419624329
for epoch: 4/5... batch: 851... G-loss:1.97986900806427.... D-loss:
0.48909667134284973
for epoch: 4/5... batch: 901... G-loss:1.4825917482376099.... D-loss:
0.49557197093963623
for epoch: 4/5... batch: 951... G-loss:1.4107017517089844.... D-loss:
0.5188608169555664
for epoch: 4/5... batch: 1001... G-loss:1.594542384147644.... D-loss:
0.4854315519332886
for epoch: 4/5... batch: 1051... G-loss:1.2983105182647705.... D-loss:
0.5350542664527893
for epoch: 4/5... batch: 1101... G-loss:1.4559698104858398.... D-loss:
0.5669841766357422
for epoch: 4/5... batch: 1151... G-loss:1.7074921131134033.... D-loss:
0.5058666467666626
for epoch: 4/5... batch: 1201... G-loss:1.5018738508224487.... D-loss:
0.5176024436950684
```

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for epoch: 4/5... batch: 1251... G-loss:1.551271677017212.... D-loss:
0.5126476287841797
for epoch: 4/5... batch: 1301... G-loss:1.5583714246749878.... D-loss:
0.4932706952095032
for epoch: 4/5... batch: 1351... G-loss:1.3716853857040405.... D-loss:
0.5502644777297974
for epoch: 4/5... batch: 1401... G-loss:1.5301686525344849.... D-loss:
0.5524863004684448
for epoch: 4/5... batch: 1451... G-loss:1.5511152744293213.... D-loss:
0.543459415435791
for epoch: 4/5... batch: 1501... G-loss:1.4957072734832764.... D-loss:
0.6009807586669922
for epoch: 4/5... batch: 1551... G-loss:1.3270671367645264.... D-loss:
0.6752053499221802

```

epoch 4



```

for epoch: 5/5... batch: 1... G-loss:1.4892628192901611.... D-loss:
0.5407747030258179
for epoch: 5/5... batch: 51... G-loss:1.6097348928451538.... D-loss:
0.518332302570343
for epoch: 5/5... batch: 101... G-loss:1.4621796607971191.... D-loss:
0.5273553133010864
for epoch: 5/5... batch: 151... G-loss:1.4143694639205933.... D-loss:
0.5952931642532349
for epoch: 5/5... batch: 201... G-loss:1.5927742719650269.... D-loss:
0.5183781385421753
for epoch: 5/5... batch: 251... G-loss:1.6489955186843872.... D-loss:
0.5674746036529541
for epoch: 5/5... batch: 301... G-loss:1.3110133409500122.... D-loss:
0.5422027111053467
for epoch: 5/5... batch: 351... G-loss:1.665252447128296.... D-loss:
0.45523518323898315
for epoch: 5/5... batch: 401... G-loss:1.4464459419250488.... D-loss:

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0.5347222089767456
for epoch: 5/5... batch: 451... G-loss:1.5878548622131348.... D-loss:
0.6383662223815918
for epoch: 5/5... batch: 501... G-loss:1.5521740913391113.... D-loss:
0.5073971152305603
for epoch: 5/5... batch: 551... G-loss:1.6604983806610107.... D-loss:
0.49660760164260864
for epoch: 5/5... batch: 601... G-loss:1.4817566871643066.... D-loss:
0.510234534740448
for epoch: 5/5... batch: 651... G-loss:1.4067726135253906.... D-loss:
0.5297749042510986
for epoch: 5/5... batch: 701... G-loss:1.6886181831359863.... D-loss:
0.5241747498512268
for epoch: 5/5... batch: 751... G-loss:1.3529915809631348.... D-loss:
0.49720942974090576
for epoch: 5/5... batch: 801... G-loss:1.5371124744415283.... D-loss:
0.6141430139541626
for epoch: 5/5... batch: 851... G-loss:1.42203950881958.... D-loss:
0.5637184381484985
for epoch: 5/5... batch: 901... G-loss:1.5535948276519775.... D-loss:
0.5320996642112732
for epoch: 5/5... batch: 951... G-loss:1.4713650941848755.... D-loss:
0.5404702425003052
for epoch: 5/5... batch: 1001... G-loss:1.431523084640503.... D-loss:
0.5500479936599731
for epoch: 5/5... batch: 1051... G-loss:1.3711620569229126.... D-loss:
0.5545493960380554
for epoch: 5/5... batch: 1101... G-loss:1.5427606105804443.... D-loss:
0.5738622546195984
for epoch: 5/5... batch: 1151... G-loss:1.2314536571502686.... D-loss:
0.6100528240203857
for epoch: 5/5... batch: 1201... G-loss:1.6364684104919434.... D-loss:
0.4885697364807129
for epoch: 5/5... batch: 1251... G-loss:1.5208590030670166.... D-loss:
0.5560282468795776
for epoch: 5/5... batch: 1301... G-loss:1.551243543624878.... D-loss:
0.5280956029891968
for epoch: 5/5... batch: 1351... G-loss:1.22129487991333.... D-loss:
0.5359112024307251
for epoch: 5/5... batch: 1401... G-loss:1.7554689645767212.... D-loss:
0.5146427154541016
for epoch: 5/5... batch: 1451... G-loss:1.507041573524475.... D-loss:
0.46773236989974976
for epoch: 5/5... batch: 1501... G-loss:1.5090404748916626.... D-loss:
0.5015310049057007
for epoch: 5/5... batch: 1551... G-loss:1.5310583114624023.... D-loss:
0.543938159942627

epoch 5

